

Name: Calvin, Andrew, Dylan **Date:** 2/06/2024

Class: Engineering Design Period 3

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - Universal Lens Cap

“The Universal Lens Cap by KUVRD.” KUVRD, kuvrdcamera.com/pages/ulc .
Accessed 6 Feb. 2024.

Details

Flexible sock design, grips to fit the shape of the lens, made of rubber

Analysis

It works at fitting every lens, is durable, and should cover the lens without damaging it



Prior Design Attempt #2 - Front lens cover

Source

“Front Lens Cover | Nikon.” www.nikonusa.com,
www.nikonusa.com/en/nikon-products/product/lens-caps/front-lens-cover.html#tab-ProductDetail-ProductTabs-Overview . Accessed 6 Feb. 2024.

Details: really long, covers the whole lens barrel, drawstring to resize the cover, made of a type of cloth (softer on the inside)

Analysis: inner material seems good to not scratch the lens, and the drawstring design creates a one-size-fits-all cover, making it even more useful (and could possibly be used as a thing to latch onto to prevent losing it).

The cloth will help to stop small scratches, however if the camera is dropped it won't do much to Prior Design Attempt #3 - Extender Cap RF



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Prior Design Attempt #3 -

“Extender Cap RF.” *Canon.com*, 2024,
usa.canon.com/shop/p/extender-cap-rf?color=Black&type=New. Accessed 6 Feb.
2024.

Details: Hard plastic, no moving parts, slides on\

Analysis: The hard cover will help to stop scratches, dents, and fingerprints, however the cover has no way of securing to the camera and is therefore easy to lose or misplace.



Prior Design Attempt #4 - LC-55A 55mm Snap-on Front Lens Cap

“LC-55A 55mm Snap-on Front Lens Cap | Nikon.” *Wwww.nikonusa.com*,
www.nikonusa.com/en/nikon-products/product/lens-caps/lc-55a-55mm-snap-on-front-lens-cap.html . Accessed 6 Feb. 2024.

This camera lens cover functions very similarly to the previous one, the way it protects the lens is just a hard cover. However, this cover attaches by using springs that can be pushed inward to make the lens fit, and then press against the rim of the lens to hold it in place (like a battery).



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Prior Design Attempt #5 -Flip-Open Scope Cover for Objective Lens

“Flip-Open Scope Cover for Objective Lens.” *Butler Creek*,
www.butlercreek.com/scope-covers/flip-open-scope-cover---objective-lens/P00180.html. Accessed 6 Feb. 2024.



<https://www.amazon.com/OP-TECH-USA-Fast-Metal/dp/B0000615L8?th=1>

A flip-open cover, made of materials such as lead

It looks too easy to break by accident (hinge), but at least the material seems durable enough to protect the lens

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Works Cited

"Front Lens Cover | Nikon." *www.nikonusa.com*,

<http://www.nikonusa.com/en/nikon-products/product/lens-caps/lc-55a-55mm-snap-on-front-lens-cap.htm>. Accessed 27 February 2024.

"Extender Cap RF." *Canon.com*, 2024,

<https://www.usa.canon.com/shop/p/extender-cap-rf?color=Black&type=New>.
Accessed 27 February 2024.

"Front Lens Cover." *Nikon*,

<http://www.nikonusa.com/en/nikon-products/product/lens-caps/front-lens-cover.html#tab-ProductDetail-ProductTabs-Overview>. Accessed 27 February 2024.

"OP/TECH USA Fast Cap with Metal Ring - 58mm Black." *Amazon.com*,

<https://www.amazon.com/OP-TECH-USA-Fast-Metal/dp/B0000615L8?th=1>. Accessed
27 February 2024.

"The Universal Lens Cap by KUVRD – KUVRD." *KUVRD*, <http://kuvrdcamera.com/pages/ulc>.

Accessed 27 February 2024.